

Jin Rhee

📍 Oxford ✉ jin.rhee@sjc.ox.ac.uk ☎ 07442639381 💻 jinrhee.github.io in jin-rhee 🌐 jinrhee

Education

University of Oxford, St John's College

Oct 2022 - Jun 2026

MEng in Engineering Science

(Expected)

- Predicted First (Prelims: 78.22, Part A: 77.11)
- Study Abroad Scholarship from [Kwanjeong Educational Foundation](#) [↗](#)

Experience

Visiting Student Researcher

Daejeon, KR

Urban Robotics Lab @ KAIST

Jul 2024 - Oct 2024

- Ran and compared trajectory error of robot odometry algorithms (FAST-LIO, LIO-SAM, and LeGo-LOAM)
- Implemented motion blur filter and open-source TSDF integration to improve quality of dense colour mesh generation from LiDAR-camera data fusion ([presentation](#)) [↗](#)
- Ground-truth labeling for 2800 frames spanning 6 regions in point-wise dynamic object segmentation dataset available at [HeLiMOS Dataset](#) [↗](#)

Student Intern

Oxford, UK

Lightricity

Mar 2024 - Mar 2024

- Implemented light sensitive sleep-time adjustment for demo to showcase ultra-low power PV cell; developed code in C on Epson MCU with EPD display and lux sensor
- Student micro-internship; presented progress to engineers via presentation

Mechatronics Engineer Intern

Seoul, KR

Bear Robotics

Jun 2023 - Sep 2023

- Directly contributed to prototype development of gantry system for food bin placement; closely collaborated within small tight-knit team
- Built production code for simultaneous step-motor control; assessed mechanical repeatability and precision of gantry; optimized trajectory planning for speed and stability
- Prototyped clamps to isolate effect of rocker-bogie mechanism in robot base; experimentally confirmed previously unknown static tip-over condition of company's main robot model

Projects

QronkOxford

[GitHub](#) [↗](#)

Oxford Robotics and Additive Manufacturing Society

Oct 2022 - current

- Co-lead of team of undergraduates creating a low-cost 3D-printed 12 DOF quadruped robot; received department's Engineering Undergraduate Innovation Fund (EUIF)
- Led motor control, power supply, leg-linkage design; led tutorial sessions on Ubuntu, Git, ROS, etc. for new members annually
- Tools used: Fusion360, ROS, C++, Python

Software and System Design of a Multi-agent Aerial System

Oct 2024 - current

Third Year Project

- Led team of four in third-year project titled *Multi-drone Assisted Search and Rescue in Disaster Scenarios*
- Designed and simulated wireless comms. in multi-drone system; culminates in final report & presentation
- Tools used: NS-3, L^AT_EX

Additional Skills and Info

Languages: C, C++, Python, MATLAB

Tools: L^AT_EX, CMake, ROS(Noetic, Foxy, Humble), Linux (Ubuntu x86, arm64), CAD (Solidworks, Fusion360), 3D printing (PLA)